

Batch Code: ANP-D1838

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LAB-10

Question:

1. Create a java program to insert a record using JDBC
 2. Create a java program to select or list a record using JDBC
 3. Create a java program to update a record using JDBC
 4. Create a java program to delete a record using JDBC
-

1. INSERT A RECORD USING JDBC

Program:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;

public class Insert {

    public static void main(String[] args) {

        Scanner scr = new Scanner(System.in);

        try
        {

            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/projects", "root",
"123456");

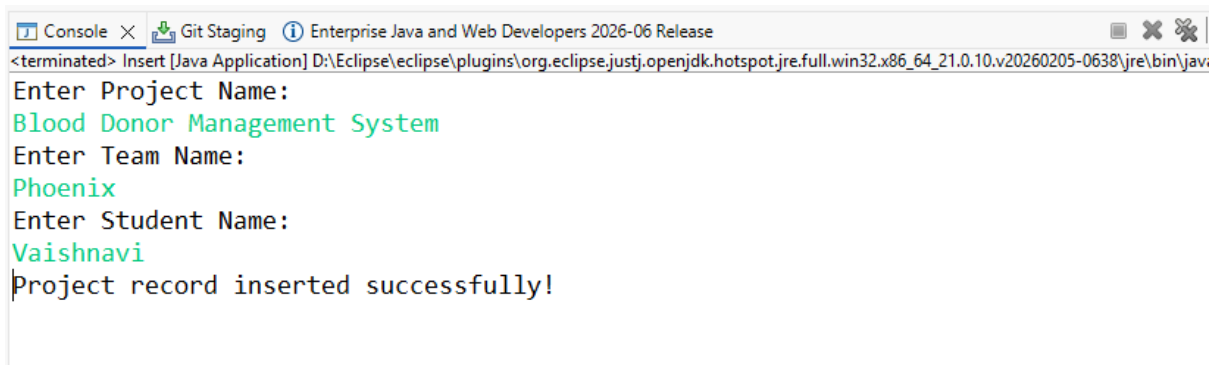
            String sql = "INSERT INTO project_details(project_name, team_name,
student_name) VALUES (?, ?, ?)";
```

```

        PreparedStatement ps = con.prepareStatement(sql);
        System.out.println("Enter Project Name: ");
        String project = scr.nextLine();
        System.out.println("Enter Team Name: ");
        String team = scr.nextLine();
        System.out.println("Enter Student Name: ");
        String student = scr.nextLine();
        ps.setString(1, project);
        ps.setString(2, team);
        ps.setString(3, student);
        ps.executeUpdate();
        System.out.println("Project record inserted successfully!");
        con.close();
    }
    catch (Exception e)
    {
        System.out.println(e);
    }
}
}

```

Output:



```

Console x Git Staging i Enterprise Java and Web Developers 2026-06 Release
<terminated> Insert [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin\jav
Enter Project Name:
Blood Donor Management System
Enter Team Name:
Phoenix
Enter Student Name:
Vaishnavi
Project record inserted successfully!

```

2. SELECT A RECORD USING JDBC

Program:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class Select {
    public static void main(String[] args) {
        try
        {

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/projects", "root", "123456");

            String sql = "SELECT * FROM project_details ORDER BY project_id";
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery(sql);

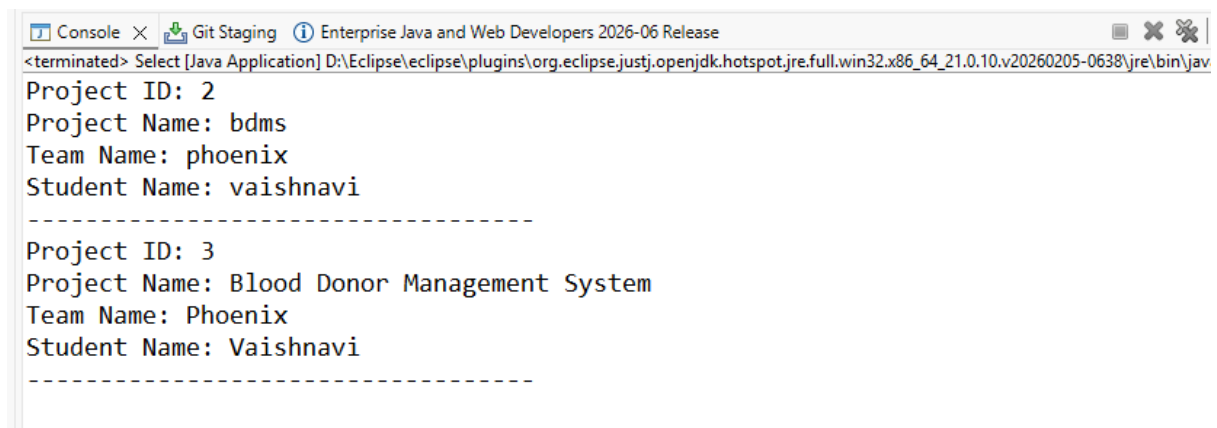
            while (rs.next()) {
                System.out.println("Project ID: " + rs.getInt("project_id"));
                System.out.println("Project Name: " + rs.getString("project_name"));
                System.out.println("Team Name: " + rs.getString("team_name"));
                System.out.println("Student Name: " + rs.getString("student_name"));
                System.out.println("-----");
            }
        }
    }
}
```

```

        con.close();
    }
    catch (Exception e)
    {
        System.out.println(e);
    }
}
}

```

Output:



```

Console x Git Staging i Enterprise Java and Web Developers 2026-06 Release
<terminated> Select [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin\jav
Project ID: 2
Project Name: bdms
Team Name: phoenix
Student Name: vaishnavi
-----
Project ID: 3
Project Name: Blood Donor Management System
Team Name: Phoenix
Student Name: Vaishnavi
-----

```

3. UPDATE A RECORD USING JDBC

Program:

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.util.Scanner;

public class Update {
    public static void main(String[] args) {

```

```
Scanner scr = new Scanner(System.in);  
try  
{  
    Connection con =  
DriverManager.getConnection("jdbc:mysql://localhost:3306/projects", "root",  
"123456");
```

```
    String sql = "UPDATE project_details SET project_name=?,  
team_name=?, student_name=? WHERE project_id=?";
```

```
    PreparedStatement ps = con.prepareStatement(sql);
```

```
    System.out.print("Enter Project ID to update: ");  
    int id = scr.nextInt();  
    scr.nextLine();
```

```
    System.out.print("Enter New Project Name: ");  
    String project = scr.nextLine();
```

```
    System.out.print("Enter New Team Name: ");  
    String team = scr.nextLine();
```

```
    System.out.print("Enter New Student Name: ");  
    String student = scr.nextLine();
```

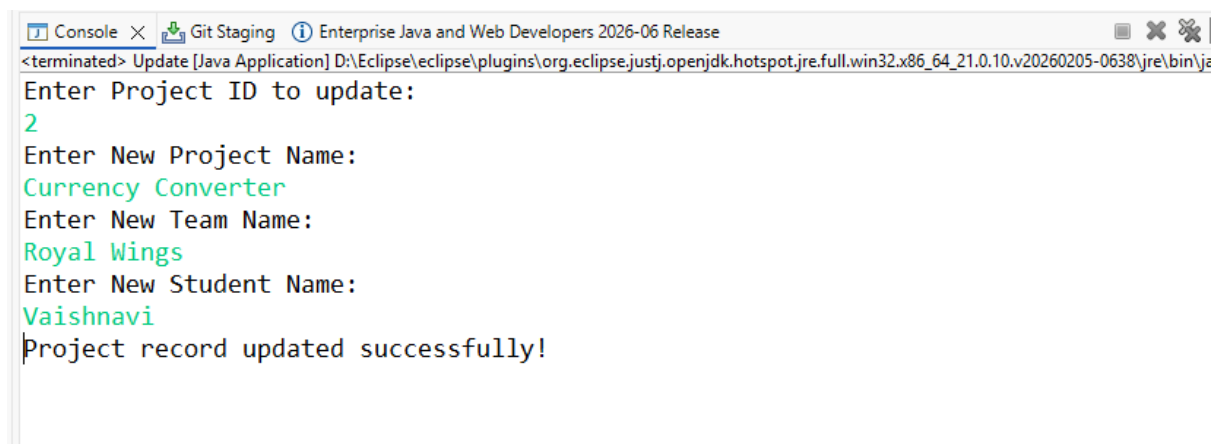
```
    ps.setString(1, project);  
    ps.setString(2, team);  
    ps.setString(3, student);  
    ps.setInt(4, id);
```

```

int rows = ps.executeUpdate();
if (rows > 0)
{
    System.out.println("Project record updated successfully!");
}
else
{
    System.out.println("No record found with Project_ID = " + id);
}
con.close();
}
catch (Exception e)
{
    System.out.println(e);
}
}
}

```

Output:



```

Console × Git Staging ⓘ Enterprise Java and Web Developers 2026-06 Release
<terminated> Update [Java Application] D:\Eclipse\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin\je
Enter Project ID to update:
2
Enter New Project Name:
Currency Converter
Enter New Team Name:
Royal Wings
Enter New Student Name:
Vaishnavi
Project record updated successfully!

```

4. DELETE A RECORD USING JDBC

Program:

```
import java.util.Scanner;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;

public class Delete {
    public static void main(String[] args) {
        Scanner scr = new Scanner(System.in);
        try
        {
            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3306/projects", "root",
"123456");

            String sql = "DELETE FROM project_details WHERE project_id=?";
            PreparedStatement ps = con.prepareStatement(sql);

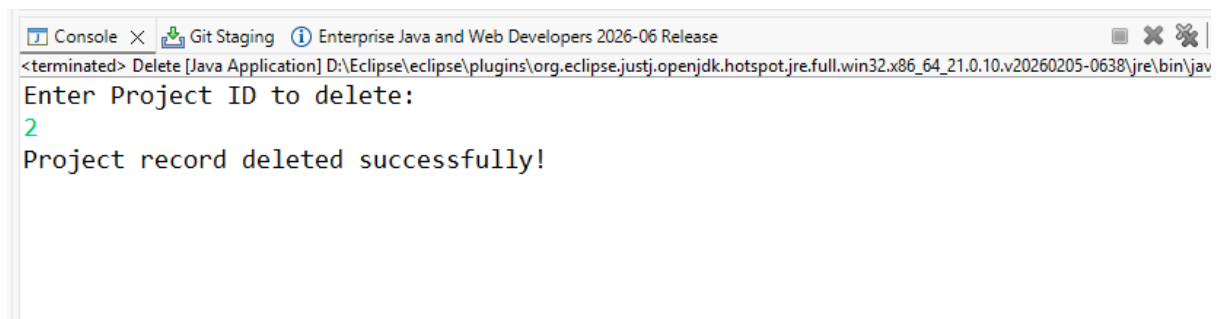
            System.out.println("Enter Project ID to delete: ");
            int id = scr.nextInt();
            ps.setInt(1, id);
            int rows = ps.executeUpdate();
            if (rows > 0)
            {
                System.out.println("Project record deleted successfully!");
            }
        }
    }
}
```

```

        else
        {
            System.out.println("No record found with Project_ID = " + id);
        }
        con.close();
    }
    catch (Exception e)
    {
        System.out.println(e);
    }
}
}

```

Output:



```

Console x Git Staging i Enterprise Java and Web Developers 2026-06 Release
<terminated> Delete [Java Application] D:\Eclipse\ eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.10.v20260205-0638\jre\bin\jav
Enter Project ID to delete:
2
Project record deleted successfully!

```

DATABASE

```

mysql> use projects
Database changed
mysql> select * from project_details;
+-----+-----+-----+-----+
| project_id | project_name | team_name | student_name |
+-----+-----+-----+-----+
| 3 | Blood Donor Management System | Phoenix | Vaishnavi |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>

```


WORKING PROCESS EXPLANATION:

INSERT PROGRAM (CREATE)

1. Connect to the database (projects).
2. Prepare an INSERT SQL statement with placeholders (?).
3. Ask the user for input (project name, team name, student name).
4. Fill those values into the prepared statement.
5. Execute the statement → a new record is added to project_details.
6. Show success message and close the connection.

SELECT PROGRAM (READ)

1. Connect to the database.
2. Prepare a SELECT * FROM project_details query.
3. Execute the query → results come back in a ResultSet.
4. Loop through each row in the result set.
5. Print the values (project_id, project_name, team_name, student_name) line by line.
6. Close the connection.

UPDATE PROGRAM (UPDATE)

1. Connect to the database.
2. Prepare an UPDATE project_details SET ... WHERE project_id=? query.
3. Ask the user for the project ID to update.
4. Ask for new values (project name, team name, student name).
5. Fill those values into the prepared statement.
6. Execute the statement → the chosen record is updated.
7. Show success message (or “No record found” if ID doesn’t exist).
8. Close the connection.

DELETE PROGRAM (DELETE)

1. Connect to the database.
 2. Prepare a DELETE FROM project_details WHERE project_id=? query.
 3. Ask the user for the project ID to delete.
 4. Fill the ID into the prepared statement.
 5. Execute the statement → the record is removed.
 6. Show success message (or “No record found” if ID doesn’t exist).
 7. Close the connection.
-